

CS 30700: Sprint Planning 3

BoilerSpace

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Sprint Overview

Distillation: For this final sprint, our primary goal is to polish, integrate, and stabilize all core BoilerSpace features while delivering remaining user facing functionality such as profiles, analytics, notifications, and reviews. We aim to ensure a cohesive, high-performance, and reliable user experience by resolving edge cases, strengthening security and privacy controls, and preparing the platform for a complete and production-ready demonstration.

Scrum Master: Sheehan Dandapat

Meeting Plan: Monday/Wednesday/Friday @ 4:30 PM in WALC 3076

Risks and Challenges:

- **Final Integration & System Stability:** As this is the final sprint, integrating all features (AI, messaging, notifications, map, profiles) without breaking existing functionality poses a major risk, especially with overlapping dependencies across teams.
- **AI Reliability & Cost Constraints:** Generating study plans and analytics using AI may produce inconsistent results, slow responses, or exceed API limits if not optimized properly.
- **Security & Privacy Enforcement:** Features like public/private profiles and admin broadcasts must strictly enforce authorization rules to prevent data leaks or unauthorized access.
- **Notification Overload & UX Clarity:** With multiple notification types (capacity alerts, content updates, broadcasts), there is a risk of overwhelming users or creating unclear UX without proper prioritization and muting controls.

Backlog (Cross out what's finished or in progress)

Functional Requirements:

Login/Profile

- ~~1. As a user, I would like to register for a BoilerSpace account so that I can save my preferences and access the full suite of features.~~
- ~~2. As a user, I would like to be able to login and manage my BoilerSpace account so that I can keep my personal data secure and up to date.~~
- ~~3. As a user, I would like to be able to verify my Purdue email by getting an OTP so that the platform remains exclusive to the campus community and prevents spam.~~
- ~~4. As a user, I would like my password to be reset if I forget it so that I can regain access to my account without creating a new one.~~
- ~~5. As a user, I would like to be able to edit my profile (display name, major, year) so~~

~~that others recognize me in sessions.~~

- ~~6. As a user, I want to block another user so that I cannot receive messages from them and their content is hidden across the platform.~~
- ~~7. As a user, I want to set my public/private profile visibility so that I control what others see.~~
- ~~8. As a user I want to set my classes for this semester, so that others know what I am studying~~
- ~~9. As a user, I want to be able to friend others based on what classes they are taking so that I can build a network of study partners for my specific coursework.~~
- ~~10. As a user, I want to update my availability or preferred study times, so that others know when I'm typically open to studying.~~
- ~~11. As a user, I want to manage my notification preferences from my profile, so that I can control how and when I'm contacted.~~
- ~~12. As a user, I want the option to upload a profile picture so others can recognize me more easily.~~
- ~~13. As a user, I want to be able to delete my account so that I can remove my personal data if I leave the university or stop using the app, and we will also have 2FA.~~

Insights

- ~~14. As a user, I want to see my total study time for this week, so that I can track my productivity.~~
- ~~15. As a user, I want to receive achievements for hitting study goals so that I stay motivated to maintain a consistent study routine.~~
- ~~16. As a user, I want AI recommendations based off my study habits so that I can improve efficiency~~
- ~~17. As a student, I would like to receive notifications for when a room is under a certain capacity so that I can travel to the location knowing I will successfully find a seat.~~
- ~~18. As a student, I want to see a clear explanation when location access is requested, so that I understand why it is needed before granting permission.~~
- ~~19. As a student, I want to receive reminders for upcoming study sessions I joined, so that I don't forget to attend.~~
- ~~20. As a student, I want to receive notifications when new content is uploaded to courses, sessions, or events that I follow so that I stay up to date without manually checking the platform.~~
- ~~21. As a student, I want to temporarily mute all notifications, so that I can focus without interruptions.~~
- ~~22. As an admin, I want to deactivate abusive accounts so that the community stays safe, if someone gets reported too many times, we will remove them from our~~

platform.

23. As an admin, I want to verify official campus organizations, so that students can trust event legitimacy.
24. As an admin, I want to remove inappropriate content (notes, comments, or events), so that the platform remains respectful.
25. As a student, I want to confirm or dispute a previous user vibe report so that the live data remains accurate and up to date
- ~~26. As a student, I want to see a timestamp of last update for a room status so I can judge how reliable the information is before I start walking~~
- ~~27. As a student, I want to filter the map for quiet zones specifically so I can find a place for deep focus rather than group work.~~
28. As a student, I want to export an AI generated study guide to PDF or markdown so I can study offline or print it out
29. As a student, I want to see a simple green or red dot next to a room name so I can instantly know if it is likely open or full
- ~~30. As a student, I want to click a single button to check in to a room so I can contribute to the live data without filling out a long form~~
- ~~31. As a student, I want to see a basic list of buildings sorted alphabetically so I can find my favorite study spot quickly~~
- ~~32. As a student, I want to see how many people are currently checked into a specific building so I can gauge the overall crowd level~~
- ~~33. As a student, I want to use a search bar to find a specific building by its name or campus abbreviation~~
- ~~34. As a student, I want my dashboard and room pages to display recent buildings I've visited, key room amenities (such as whiteboards and outlets), and the last updated time of room status so that I can quickly choose a study space that fits my needs and trust the availability information.~~
- ~~35. As a student, I want to filter my course list by the current semester so I can focus only on the classes I am taking right now~~
- ~~36. As a student, I want to see a small icon next to a building name that indicates if it has a cafe or vending machines nearby so I can choose a study spot based on snack availability~~
37. As an admin, I want automatic spam detection so less moderation work is needed
- ~~38. As an admin, I want to be able to broadcast announcements so users can receive platform updates~~
39. As an admin, I want to see platform analytics to see which features are most popular, least-used, and make data-driven decisions to improve the platform
- ~~40. As a student looking for a seat, I want to view an interactive campus map so that I can quickly see buildings and rooms.~~

- ~~41. As a student I want directions to different study locations so that I can find the fastest route to a building I haven't visited before.~~
- ~~42. As a student, I want to see a list of rooms sorted by distance from my current location so that I can find the nearest available spot.~~
- ~~43. As a student, I want to bookmark my favorite rooms so that I can quickly check them in the future.~~
- 44. As a student I would like to find people from my class to study with so that we can help each other understand course material.
- 45. As a student, I want to search for certain classes that people are studying for so that I can join a group that is working on the same assignment/project as me.
- 46. As a student, I want to search for certain clubs and if they are hosting any study get-togethers so that I can meet people with shared interests.
- ~~47. As a student, I want to comment on a session's notes so that I can ask clarifying questions or provide corrections to shared content.~~
- 48. As a student, I want to know certain exam dates for certain classes so that I can prioritize my study sessions effectively.
- 49. As a student, I want the option of dark mode so that I can study more comfortably at night.
- 50. As a student, I want to adjust the text size, so that all the content is easier to read.
- ~~51. As a student, I want to view and upload notes from study sessions for my course so that I can both review material I may have missed and contribute to the shared knowledge base.~~
- ~~52. As a student, I want to upvote/downvote certain notes so that the most helpful and accurate resources rise to the top.~~
- ~~53. As a student, I want to rate and post reviews on study areas so others can find the best spots~~
- ~~54. As a student, I want to filter rooms based off amenities (whiteboards, outlets) so that I can find a workspace that fits my equipment needs.~~

AI Assistant

- ~~55. As a student, I want to ask AI assistant questions about the aggregated study notes for my course so that I can find specific explanations without reading every document.~~
- 56. As a student, I want the AI assistant to show citations that link back to the exact note or page it used, so that answers are traceable and verifiable.
- ~~57. As a student, I want the AI assistant to generate concise study guides from my notes and highlight key topics, so I can review efficiently and know what to focus on.~~
- 58. As a student, I want the AI to suggest practice questions without giving full

~~solutions, so that I can test my understanding.~~

59. As a student, I want to bookmark helpful AI responses, so that I can easily return to them later.

~~60. As a student, I want the AI to help create a study plan and schedule so I can organize my time efficiently~~

61. As a student, I want AI to suggest external resources like textbooks or videos to improve my understanding of certain topics

Clubs / Organizations

~~62. As a club organizer, I want to post events with time and location so that students can discover them easily.~~

63. As a club organizer, I want to tag events by category (callout, workshop, social, fundraiser) so that students can filter them.

~~64. As a club organizer, I want to create an event with a title, description, date, time, and campus location, so that students clearly understand what the event is.~~

65. As a club organizer, I want to manage who can create events under my club so that only authorized members can post.

66. As a club organizer, I want to attach external links (sign-up forms, Discord, social media) so that students can engage beyond the event.

67. As a club organizer, I want to see how many students plan to attend so that I can plan resources.

~~68. As a club organizer, I want to post announcements or updates related to events, so that attendees stay informed.~~

69. As a club member I want to RSVP to events so that the organizers know to expect me.

70. As a club organizer, I want to restrict even visibility to certain audiences (member-only or public), so that events are targeted.

~~71. As a club organizer, I want to create and manage a club profile, so that students can learn about the organization.~~

~~72. As a club organizer, I want to assign different roles to club members (e.g., admin, officer) so that responsibilities are clear.~~

~~73. As a club organizer, I want to approve or remove club members, so that organizations stay moderated.~~

~~74. As a club organizer, I want to schedule recurring events so members can plan attendance in advance~~

75. As a club member, I want to see files or slides from past events to stay up to date

76. As a club organizer, I want to see past attendance statistics so I can evaluate engagement

Communication

- 77. As a student, I want threaded discussions for session pages, so that conversations are organized. (if time allows)
- 78. As a student, I want to reply to specific messages within a session, so that side conversations stay contained within the main discussion thread.
- 79. As a student, I want to receive notifications when someone replies to a thread I participated in, so that I can stay engaged without monitoring the entire chat.
- 80. As a student, I want to mute specific threads within a session, so that I can focus on conversations that matter to me.
- 81. As a student, I want to see a timestamp on messages in session discussions, so that conversations have context.
- 82. As a student, I want to search within a session's discussion threads, so that I can find specific information.
- ~~83. As a student, I want to react to messages with simple reactions, so that I can acknowledge messages without replying.~~
- 84. As a student, I want to delete messages I accidentally sent so others see a deletion notice, and I want the option to send auto-disappearing messages that vanish after a set time so sensitive content doesn't linger.
- ~~85. As a user, I want to send direct messages to another user and view my conversation history so that I can coordinate study plans privately.~~
- ~~86. As a user, I want to view my own and other users' public profiles so that I can learn more about potential study partners before connecting with them.~~
- ~~87. As a user, I want to see typing indicators, read receipts, and online/offline presence in my DMs so I know when others are active and have seen my messages.~~
- ~~88. As a user, I want to approve DM requests from people I'm not friends with so strangers can't message me without permission.~~
- ~~89. As a user, I want to search through all my messages globally and within a specific conversation so I can find anything without scrolling.~~

Non-Functional Requirements

Architecture and Performance

BoilerSpace will use a client server architecture based on the MERN stack (MongoDB, Express, Node.js) to support maintainability and future growth. The frontend will be a responsive web application built with React and Tailwind CSS, while the backend will provide a REST API built with Node.js and Express to manager user accounts, room availability, study sessions, file uploads, and AI-assisted queries. This separation allows system components to be developed and updated independently. The system will be designed to provide responsive performance under typical usage. Common actions such as viewing room availability, browsing study sessions, and loading the campus map should be completed within 1 second. More resource intensive actions including operations with the AI Knowledge Chatbot should be completed within

3 seconds. The application is expected to support approximately 300 concurrent users without noticeable performance degradation. Authentication and authorization will be implemented using Passport.js within our Express backend. User passwords will be salted and hashed using bcrypt (with a work factor of 10) before storage in MongoDB. To secure the REST API, we utilize the passport-jwt strategy, requiring a valid, signed JSON Web Token or JWT in the authorization header for all protected endpoints. This stateless authentication method prevents Cross-Site Request Forgery and ensures data scraping is blocked.

Crowdsourced Data Integrity (Security)

BoilerSpace relies on crowdsourced check-ins to provide live room availability and study environment indicators. Room data should be based on combined input from multiple users, while still allowing recent individual check-ins to influence results when limited data is available. The system should account for users leaving rooms by gradually reducing the impact of older check-ins over time, ensuring that availability indicators stay aligned with real campus conditions.

Usability

The BoilerSpace interface emphasizes clear organization to help users efficiently discover study spaces, sessions, and campus events. The interactive campus map enables users to explore locations, view ability, and access relevant information with minimal effort. Study sessions, courses, and club events are structured with filtering and search options to make content easy to find and sort. The AI chatbot is designed to be easily accessible within the application, allowing users to ask questions and review course related content without disrupting their workflow. Users should also be easily able to view past notes and materials shared by other students or generated through the AI assistant. Profile features such as class selection and visibility controls are streamlined so users can manage how they are represented to others. The application will provide a consistent and usable experience across devices, screen sizes, and modern web browsers.

Hosting/Deployment

To maintain continuous delivery and scalability, the application will use a split deployment strategy. The frontend will be hosted on Vercel, which provides automatic deployments from our GitHub repo and a global CDN for fast load times. The backend will be deployed on Render. This way we can implement our crowdsourcing and messaging features, which require a long-running server process.

Current Sprint Detail

User Story 53: Study Area Reviews

Story: As a student, I want to rate and post reviews on study areas so others can find the best spots.

Tasks (Jed)

1. **Design review and rating database schema — 2.0 hrs** Create a new **Review** schema that includes fields for a numerical rating (1-5 stars), a text comment, a timestamp, and references to both the **User** (author) and the **Room** (location).
2. **Develop CRUD API endpoints for reviews — 3.0 hrs** Implement a **POST /reviews** endpoint to handle new submissions and a **GET /reviews/:roomId** endpoint that retrieves all reviews for a specific location. Include logic to calculate and return the average rating for the room.
3. **Build review submission form and star-rating UI — 2.5 hrs** Create a React component that allows users to select a star rating and type a text review. Include basic front-end validation to ensure a rating is selected before submission.
4. **Implement reviews display list on room detail page — 1.5 hrs** Develop a scrollable list component that renders existing reviews, showing the user's name, their rating, and the date of the post. Add a summary header displaying the room's total average score.
5. **Testing and validation coverage — 1.0 hr** Write unit tests to ensure ratings are within the 1-5 range and verify that the review list correctly displays data associated only with the selected room.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I am on a room's detail page, when I select a star rating and enter a comment in the review section, then my review is saved and appears at the top of the review list.
- Given a room has multiple user ratings, when I view that room's detail page, then I see an accurately calculated average star rating (e.g., 4.2 / 5 stars).
- Given I am viewing a list of reviews, when a new review is posted by another user, then the list updates to include the most recent feedback.

User Story 27: Quiet Zone Filtering

Story: As a student, I want to filter the map for quiet zones specifically so I can find a place for deep focus rather than group work.

Tasks (Jed)

1. **Refactor map to remove hardcoded noise levels and colors — 3.0 hrs** Locate the existing map marker logic and remove the placeholder/static noise values and hex codes. Refactor the component to use a dynamic color-mapping utility that assigns colors based on the **noiseLevel** field returned by the API.

2. **Update Room schema and backend API — 2.0 hrs** Modify the database schema to include a `noiseClassification` field (e.g., "Quiet", "Moderate", "Collaborative"). Update the `GET /rooms` endpoint to ensure this data is serialized and sent to the frontend.
3. **Implement map filtering logic for quiet zones — 2.5 hrs** Develop the state management logic within the Map view to filter the array of room objects. Ensure that when the filter is active, only rooms with a "Quiet" classification are passed to the rendering engine.
4. **Build the UI filter toggle component — 1.5 hrs** Create a "Quiet Zones Only" toggle button or checkbox on the Map interface. Style it to match the BoilerSpace design system and ensure it provides clear visual feedback when active.
5. **Testing and validation coverage — 1.0 hr** Write unit tests to verify that the filter correctly excludes non-quiet rooms and ensure that the map no longer relies on hardcoded color strings for its markers.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I am looking at the campus map, when I toggle the "Quiet Zones" filter to ON, then only rooms marked as quiet study areas remain visible.
- Given the map is loaded, when I view room markers, then their colors are determined by real-time database values rather than random hardcoded defaults.
- Given the "Quiet Zones" filter is active, when I zoom or pan the map, then the filter remains persistent and continues to hide collaborative spaces.

User Story 38: Admin Broadcast Announcements

Story: As an admin, I want to be able to broadcast announcements so users can receive platform updates.

Tasks (Jed)

1. **Develop Announcement schema and POST /broadcast endpoint — 3.0 hrs** Update `Announcement` schema to include fields for title, body, priority level (e.g., info, warning, alert), and expiration date. Build a protected backend route that only allows users with "Admin" privileges to successfully post new broadcasts.
2. **Build Admin Broadcast Dashboard UI — 3.0 hrs** Create a dedicated management page for admins to draft announcements. This includes a form with validation, a "Live Preview" of the announcement banner, and a list of active/expired broadcasts with the ability to delete or archive them.
3. **Implement Global Announcement Banner in React — 2.0 hrs** Develop a high-visibility banner component that sits at the top of the main layout. It must fetch the most recent active announcement on app load and display it across all pages (Map, Room Details, etc.) until the session ends or it is dismissed.
4. **Add dismissible state and local storage logic — 1.0 hr** Implement logic that allows users to close a broadcast. Use `localStorage` to ensure that once a student dismisses a specific announcement ID, it does not reappear for them during future sessions until a new broadcast is posted.

5. **Testing and validation coverage — 1.0 hr** Write tests to verify that the broadcast endpoint rejects non-admin requests and ensure the React banner correctly renders different styles based on the "priority level" (e.g., red for alerts, blue for info).

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I am logged in as an Admin, when I submit the broadcast form, then a new announcement is saved to the database and becomes immediately available to all users.
- Given there is an active broadcast, when a student opens the BoilerSpace app, then they see a banner at the top of the screen containing the admin's message.
- Given I have read an announcement, when I click the "X" to dismiss it, then the banner disappears and does not return when I refresh the page.

User Story 86: Public Profile Views

Story: As a user, I want to view my own and other users public profiles so I can learn more about study partners before connecting with them.

Tasks (Sheehan)

1. **Design and implement Profile UI component - 4.0 hrs** Create a React component displaying user info (avatar, bio, major, courses). This needs distinct states for "own profile" (which includes edit buttons) and "other user profile" (read-only, with a button to initiate a connection).
2. **Develop GET /profile/:userId endpoint - 3.0 hrs** Create a backend route to fetch user profile data. The logic must securely strip out sensitive or private data if the requested profile does not belong to the requesting user.
3. **Implement "Connect" button logic - 2.0 hrs** Add frontend state management and backend integration to send a connection request from the profile view, updating the UI to show a "Pending" state once clicked.
4. **Testing and validation coverage - 1.0 hr** Write tests ensuring private data is not leaked in the payload when viewing someone else's profile, and verify that the UI renders gracefully if certain profile fields are left blank.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I am logged into the platform, when I navigate to my own profile, then I see all my information and the options to edit my details.
- Given I am looking at a potential study partner's profile, when the page loads, then I only see the information they have made publicly available.
- Given I am viewing another user's profile, when I click "Connect," then a request is sent to them and the button updates to reflect the pending status.

User Story 7: Profile Visibility Toggle

Story: As a user, I want to set my public/private profile visibility so that I control what others see.

Tasks (Sheehan)

1. **Update User schema and DB - 2.0 hr** Add an `isPublic` boolean field to the User model in the database, defaulting to `true` for new accounts.
2. **Build Settings UI toggle - 3.0 hrs** Create a privacy section within the account settings page. Implement a toggle switch for "Public Profile" with clear helper text explaining what information is hidden when set to private.
3. **Implement PUT /user/visibility endpoint - 1.5 hrs** Create a protected endpoint to handle the toggle state change, ensuring it securely updates the user's document in the database and returns the updated state.
4. **Update search and directory logic - 2.0 hrs** Modify backend queries for the study partner directory so that users with `isPublic` set to `false` are excluded from search results. Update the profile fetch endpoint to return a 403 or "Private Profile" placeholder if a private profile is accessed directly via URL.
5. **Testing and validation coverage - 1.5 hrs** Write integration tests to confirm private profiles cannot be queried in the directory and that unauthorized users cannot bypass the UI to view private profile data.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I am on my account settings page, when I toggle my visibility to "Private," then my profile is immediately updated in the database.
- Given my profile is set to Private, when another user attempts to navigate directly to my profile URL, then they see a message indicating the profile is private.
- Given my profile is set to Public, when another user searches for study partners in my major, then my profile appears in their search results.

User Story 14: Weekly Study Time Analytics

Story: As a user, I want to see my total study time for this week, so that I can track my productivity.

Tasks (Sheehan)

1. **Develop Study Session schema & tracking logic - 3.0 hrs** Create database models to record the start, end, and total duration of individual study sessions tied to a specific user ID.
2. **Build GET /analytics/weekly endpoint - 3.5 hrs** Create backend logic to query and aggregate the user's study session durations. The query must filter for sessions that occurred between the current week's Monday at 12:00 AM and Sunday at 11:59 PM.
3. **Implement Dashboard Analytics UI - 2.5 hrs** Build a frontend widget for the main dashboard that fetches the aggregated weekly data. Display the total hours

and minutes in a clean UI, potentially utilizing a progress bar if the user has set a weekly goal.

4. **Testing and validation coverage - 1.0 hr** Write tests to ensure time zone calculations are accurate (so Monday midnight aligns with the user's locale) and that aggregations perfectly match the raw session data in the database.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I have completed study sessions this week, when I navigate to my dashboard, then I see my total hours studied accurately calculated and displayed.
- Given it is a new week (Monday @12:00 AM), when I check my dashboard, then the weekly study time counter is reset to zero.
- Given I am viewing my dashboard, when I finish logging a new study session, then my weekly total updates immediately to reflect the newly added time.

User Story 20: Notifications

Story: As a student, I want to receive notifications when new content is uploaded to courses, sessions, or events that I follow so that I stay up to date without manually checking the platform.

Tasks (Bidit)

1. **Create Notification Schema – 2.0 hours:**

Update database to include Notification schema and link users with courses/events that they follow.

2. **Update Backend to Detect Uploads and Posts – 3.0 hours:**

Update backend to detect new uploads or event creations and push a message to all users that follow the event.

3. **Create Notification API and Frontend Pop-up – 2.5 hours:**

Build an endpoint that gets all unread notifications and create a frontend popup component to display them

4. **Integrate Email Notification Service – 1.0 hours:**

Use a mailer service to send external alerts based on a users notification preferences

5. **Testing and Validation Coverage – 1.5 hours:**

Test notification sending and reading.

Estimated total: 10.0 hours

Acceptance Criteria:

- Given a new file is uploaded to a course I added, when the upload is complete, then I receive a notification.
- Given I have followed an event, when event details are created or updates, then I receive an alert.
- Given I click on a notification, then I am redirected to the content that the notification was for.

User Story 21: Mute Notifications

Story: As a student, I want to temporarily mute all notifications, so that I can focus without interruptions.

Tasks (Bidit)

1. Add User Privacy Settings – 2.0 hours:

Update User schema to include an `is_muted` variable and a `mute_duration` timestamp for focus sessions

2. Build Mute UI – 2.0 hours:

Create a “Focus Mode” toggle in the user’s profile settings that allows users to set a duration for muted notifications

3. Implement Notification Muting Logic – 4.5 hours:

Modify backend notification logic to check user’s mute preferences before sending any notifications, and automatically re-enables notifications once a user’s `mute_duration` expires.

4. Testing and Validation Coverage – 1.5 hours:

Test that notifications are queued but not displayed during mute mode and appear correctly once mute mode expires.

Estimated total: 10.0 hours

Acceptance Criteria:

- Given I toggle “Focus Mode” to be ON, then all notification alerts and sounds are suppressed.
- Given I set a mute timer, when the time expires, then my notification status reverts back to active.
- Given I muted notifications, when I check my inbox later, then I can still see the all the notifications I missed during “Focus Mode”.

User Story 60: AI Study Plan

Story: As a student, I want the AI to help create a study plan and schedule so I can organize my time efficiently.

Tasks (Bidit)

1. Develop Study Plan Schema – 2.5 hours:

Design StudyPlan schema in backend that takes user data, such as course load, preferred study hours, and deadlines, to feed to AI model.

2. Implement Gemini Scheduling Logic – 3.5 hours

Develop backend service that sends user data to the Gemini API using a prompt designed to output a structured JSON calendar format.

3. Build Interactive Calendar UI – 2.5 hours:

Create a view where the AI-generated plan is displayed as a weekly calendar grid or something else.

4. Testing and Validation Coverage – 1.5 hours:

Test that study plan is properly generated using user data and is properly saved in user’s history.

Estimated total: 10.0 hours

Acceptance Criteria:

- Given I provide my exam date, when I click "Generate Study Guide," then I receive a comprehensive overview of the material.
- Given a generated plan, when I find a conflict, then I can manually edit specific sections of the schedule.
- Given I have an existing personal calendar, when the study plan is created, then it allocates study blocks around my existing sessions without overlapping.

User Story 87: Real-Time DM Enhancements

Story: As a user, I want to see typing indicators, read receipts, and online/offline presence in my DMs so I know when others are active and have seen my messages.

Tasks (Aagam)

1. Add readAt field to Message schema — 2.0 hrs Emit a read event to all conversation participants when the recipient opens the conversation and update the message's readAt timestamp in the database.
2. Build typing indicator via Socket.io — 2.0 hrs Emit a typing event on keypress with a 2 second debounce so the indicator disappears automatically after the user stops typing.
3. Track online/offline presence — 2.0 hrs Store connected userIds in a server-side socket Map and emit presence events on connect and disconnect to all relevant clients.
4. Update chat UI — 2.5 hrs Show "Seen" under the last message, "X is typing..." in the chat footer, and a green or grey dot next to each user in the DM inbox.
5. Testing and validation coverage — 1.5 hrs

Estimated total: 10.0 hrs

Acceptance Criteria:

- Given I am typing, when the other user has the chat open, then they see "X is typing..."
- Given I send a message, when the recipient opens the chat, then I see "Seen" below my message
- Given a user goes offline, when their socket disconnects, then their green dot turns grey in my inbox
- Given I open an old conversation, when the history loads, then each message correctly shows "Seen" or "Delivered"

User Story 88: DM Conversation Requests

Story: As a user, I want to approve DM requests from people I'm not friends with so strangers can't message me without permission.

Tasks (Aagam)

1. Add status field to Conversation schema — 2.0 hrs Add a status field with values pending, accepted, and rejected, defaulting to pending for any conversation started between non-friends.
2. Build accept and reject endpoints — 2.0 hrs Implement POST /conversations/:id/accept and POST /conversations/:id/reject and enforce that only the recipient can action the request.
3. Queue messages until accepted — 2.5 hrs Prevent socket message delivery to the recipient until the conversation is accepted, then flush all queued messages at once on approval.
4. Build Message Requests tab in DM inbox — 2.0 hrs Add a separate Requests tab showing pending conversations with the sender's name, profile picture, message preview, and accept/reject buttons.
5. Testing and validation coverage — 1.5 hrs

Estimated total: 10.0 hrs

Acceptance Criteria:

- Given a non-friend messages me, when I open my inbox, then their message appears in a "Requests" tab not my main inbox
- Given I accept a request, when it is approved, then the conversation moves to my main inbox and their messages appear
- Given I reject a request, when they try to message me again, then they are blocked from starting a new conversation
- Given two users are already friends, when one messages the other, then it goes straight to the main inbox with no request step

User Story 89: DM Message Search

Story: As a user, I want to search through all my messages globally and within a specific conversation so I can find anything without scrolling.

Tasks (Aagam)

1. Add MongoDB text index on Message content field — 0.5 hrs Create a text index on the content field of the Message schema to enable full-text search queries.
2. Build per-conversation search endpoint — 2.0 hrs Implement GET /conversations/:id/search?q=&page= returning paginated results with one message of context above and below each match.
3. Build global search endpoint — 2.5 hrs Implement GET /messages/search?q=&page= that searches across all conversations the user is in and returns which conversation each match belongs to.
4. Build per-conversation search UI — 2.0 hrs Add a search bar inside the chat window that highlights matched keywords and includes a "Jump to message" button that scrolls to and briefly flashes the original message.
5. Build global search UI — 1.0 hr Add a search bar in the DM inbox header that renders results grouped by conversation under each user's name as a section header.
6. Testing and validation coverage — 1.0 hr

Estimated total: 10.0 hrs

Acceptance Criteria:

- Given I search a keyword in my inbox, when results load, then I see matches grouped by conversation with the keyword highlighted
- Given I click a result, when the chat opens, then it jumps to that message and briefly highlights it
- Given I search inside a specific conversation, when results load, then I see each match with one message of context around it (the above message and below, if they exist)
- Given my search returns nothing, when the query completes, then I see "No messages found for X"

User Story 72: Club Role Management

Story:

As a club organizer, I want to assign roles and create custom positions for members so that responsibilities are clearly defined and access is controlled for each club.

Tasks (Nathan)

- **Update ClubMember schema with role and position fields - 1.5 hrs**
Add a role field (member, officer, admin) and a position field to the ClubMember model, with default values and support for custom positions per club.
- **Implement custom positions per club - 2.0 hrs**
Allow club organizers to define a set of allowed positions (e.g., Event Lead, Treasurer) unique to each club and store them in the database.
- **Implement role-based access control logic - 1.5 hrs**
Create middleware to enforce permissions based on both role and position, ensuring users can only perform actions allowed by their assigned role.
- **Implement PATCH /clubs/:clubId/members/:userId/role endpoint - 1.5 hrs**
Create a backend endpoint that allows authorized users to update a member's role and/or position.
- **Implement endpoints for managing custom positions - 1.5 hrs**
Add routes to create, update, and delete custom positions for a club, restricted to admins.
- **Build role & position management UI - 1.5 hrs**
Add UI components in the club dashboard to view members, assign roles, and assign or create positions from a list of allowed positions.
- **Prevent unauthorized role/position changes - 0.5 hrs**
Ensure users cannot assign roles or positions higher than their own permissions and cannot assign invalid positions.

Estimated Total: 10 hrs (you can trim if needed)

Acceptance Criteria

- Given I am a club admin, when I create custom positions for my club, then those positions are available for assignment to members.
- Given I assign a role and position to a member, when the action is completed, then both values are saved and correctly reflected in the database.
- Given I am not an admin, I do not have the option to change my own role
- Given custom positions are created, when I view the member list, then each member's role and position are displayed clearly.
- Given I assign a position to a member, when I select from the UI, then I can only choose from the list of valid positions defined for that club.

User Story 83: Message Reactions

Story: As a student, I want to react to messages with simple reactions so that I can acknowledge messages without replying.

Tasks (Nathan)

- **Update Message schema with reactions field - 1.5 hrs**
Add a reactions array to the Message model to store userId and reaction type.
- **Implement POST /messages/:id/reactions endpoint - 2.0 hrs**
Create a backend endpoint to add, update, or remove a reaction from a message, ensuring one reaction per user per message.
- **Implement reaction toggle logic - 1.5 hrs**
Allow users to toggle reactions on and off, replacing existing reactions if a different type is selected.
- **Build reaction UI components - 2.0 hrs**
Add clickable emoji reactions to each message in the UI and display counts for each reaction type.
- **Display aggregated reaction counts - 2.0 hrs**
Show total counts per reaction type and update the UI dynamically after a user reacts.
- **Testing and validation coverage - 1.0 hr**
Test reaction creation, updates, and removal, and ensure that duplicate reactions from the same user are prevented.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I click a reaction on a message, when I select an emoji, then my reaction is saved and displayed on the message.
- Given I already reacted to a message, when I choose a different reaction, then my previous reaction is replaced.
- Given a message has reactions, when it is displayed, then all reaction counts are shown correctly and updated in real time.
- Given I reacted to a message, if I exit the conversation and come back to it, the reaction is still there

User Story 74: Recurring Events + Club Calendar View

Story: As a club organizer, I want to schedule recurring events and view them in a calendar so that members can plan attendance in advance and easily see all events for the month

Tasks (Nathan)

- **Extend Event schema with recurrence fields - 1.5 hrs**

Add a recurrence object to the Event model including fields such as type (none, weekly, monthly), interval, end date, and a recurrenceGroupId to link related events in a series.

- **Implement recurring event generation logic - 2.5 hrs**

Modify event creation logic to generate multiple future event instances based on recurrence rules (e.g., weekly or monthly) and store them as separate events linked by a shared recurrenceGroupId.

- **Implement edit logic for recurring events - 2.0 hrs**

Add support for editing a single event, all events in a series, or future events only, ensuring correct updates are applied across related events when necessary.

- **Build backend endpoints for recurring events - 1.5 hrs**

Create or update endpoints to handle creation, retrieval, updating, and deletion of recurring event series, including filtering by club and recurrence group.

- **Implement calendar view UI on club page - 2.0 hrs**

Build a calendar interface (monthly view) within each club page that visually displays all events, including recurring events, in their correct dates.

- **Integrate event display with calendar - 0.5 hrs**

Ensure events appear correctly on the calendar and are clickable to view details, including support for recurring event instances.

- **Testing and validation coverage - 0.5 hrs**

Test recurrence generation, editing behaviors, and calendar rendering to ensure events display correctly across different scenarios and edge cases.

Estimated total: 10.0 hrs

Acceptance Criteria

- Given I create a recurring event with a weekly or monthly frequency, when I submit the event, then multiple future instances of the event are automatically generated and stored in the database.
- Given I view the club page, when I open the calendar view, then I see all events for the current month displayed in their correct dates, including recurring events.
- Given I edit a recurring event, when I choose to apply changes to a single event, then only that specific instance is updated.
- Given I edit a recurring event, when I choose to apply changes to all events in the series, then all related events are updated accordingly.
- Given I delete a recurring event, when I confirm deletion, then I am prompted to delete either a single occurrence or the entire series, and the system updates accordingly.
- Given I am on the club page, when I view the calendar, then I can click on any

event to see its full details.

- Given events are scheduled, when multiple events occur on the same day, then all events are displayed and accessible within that date on the calendar.

User Story 11: Notification Preferences

Story: As a user, I want to manage my notification preferences from my profile, so that I can control how and when I'm contacted.

Tasks (Rohan)

- **Update User Schema & Database Defaults — 2.5 hrs** Modify the `User` model to include a nested `notificationPreferences` object with boolean flags for session reminders, messages, events, organization updates, and a global mute. Ensure new accounts default to active notifications.
- **Develop Notification Settings API — 3.0 hrs** Create the `PUT /users/:id/notifications/preferences` endpoint within the `SocialController` (or `UserController`). Implement authorization checks to ensure users can only modify their own settings.
- **Integrate Logic with Notification Service — 3.0 hrs** Update the backend `NotificationService` to actively check these user preference flags before emitting Socket.io real-time alerts or executing background push jobs.
- **Testing and State Validation — 2.0 hrs** Write integration tests to verify that setting a preference flag to `false` successfully intercepts and blocks the corresponding notification payload from firing.

Estimated total: 10.5 hrs

Acceptance Criteria

- Given I navigate to my Settings page, when I open "Notification Preferences," then I see individual toggles for sessions, messages, and events.
 - Given I toggle "Session Reminders" to off, when my joined study session is about to begin, then I do not receive a platform alert.
 - Given I want to pause all alerts, when I toggle "Mute All Notifications," then all underlying notification categories are disabled until I manually turn them back on.
 - Given I update my preferences on the web client, when I refresh the page, then my settings are saved and accurately reflected in the UI.
-

User Story 48: Exam Dates Tracking

Story: As a student, I want to know certain exam dates for certain classes so that I can prioritize my study sessions effectively.

Tasks (Rohan)

- **Update Course Schema with Exam Schedules — 2.5 hrs** Modify the **Course** database model to include an array of **Exam** subdocuments. Each subdocument should contain a title (e.g., "Midterm 1"), date/time, and an optional location string.
- **Build Exam Data API Endpoints — 3.5 hrs** Develop the **GET /courses/:id/exams** endpoint to fetch upcoming test dates. Implement a protected **POST /courses/:id/exams** endpoint for authorized users or admins to crowdsource and add new dates to the syllabus.
- **Integrate Exam Dates into Course UI — 3.0 hrs** Update the "Course Page" frontend component to parse the exam array and render a chronological list of upcoming exams at the top of the course dashboard.
- **Testing and Timezone Handling — 1.5 hrs** Write tests to ensure timezone conversions display accurately for the local client and implement a filter to automatically archive or hide exam dates that have already passed.

Estimated total: 10.5 hrs

Acceptance Criteria

- Given I am on a specific Course Page, when I view the dashboard, then I see a dedicated section displaying the dates for upcoming midterms or finals.
- Given I am planning to host a study session, when I look at the course details, then I can clearly see the next exam date to help me schedule the meeting appropriately.
- Given an exam date has passed, when I view the course page the following day, then that specific exam is no longer highlighted as an upcoming priority.
- Given I am browsing my enrolled classes, when I open a course with an exam this week, then the date is prominently displayed to help me prioritize my workload.

User Story 59: Bookmark AI Responses

Story: As a student, I want to bookmark helpful AI responses, so that I can easily return to them

later.

Tasks (Rohan)

- **Design Bookmark Database Schema — 2.5 hrs** Create an `AIBookmark` schema that stores a `userId`, the original `promptString`, the `aiResponseText`, and a timestamp. Add a database index on the `userId` for fast querying.
- **Implement Bookmark CRUD Endpoints — 3.5 hrs** Develop `POST /users/:id/bookmarks/ai` to save a generated response to the database, and `GET /users/:id/bookmarks/ai` to retrieve a user's saved list within the `ResourceController`.
- **Build Frontend Bookmark UI Integration — 3.0 hrs** Add a clickable "Bookmark" icon to the AI chat bubbles in the `AIDChatAssistant` component. Create a dedicated "Bookmarked Resources" view in the Courses/Library section to render the saved items.
- **Testing and Data Retrieval Validation — 1.5 hrs** Write unit tests to ensure users can only fetch their own bookmarks. Implement tests for a `DELETE` route to verify users can successfully remove items from their saved list.

Estimated total: 10.5 hrs

Acceptance Criteria

- Given the AI assistant provides a helpful explanation, when I click the bookmark icon on the chat bubble, then the message is saved to my account.
- Given I have saved multiple AI responses, when I navigate to my Bookmarked Resources page, then I can view a scrollable list of all my saved explanations.
- Given I am viewing my bookmarked AI responses, when I click on a specific item, then I can read both the full question I asked and the AI's generated answer.
- Given I no longer need a saved response, when I click the "remove bookmark" icon, then the response is permanently deleted from my saved list and disappears from the UI.

Summary of Assignments

- **Aagam:**
 - User Story 87: Real-Time DM Enhancements
 - User Story 88: DM Conversation Requests
 - User Story 89: DM Message Search
- **Bidit:**
 - User Story 20: Receive Notifications
 - User Story 21: Mute Notifications
 - User Story 60: AI Study Plan
- **Jed:**
 - User Story 53: Study Area Reviews
 - User Story 27: Quiet Zone Filtering
 - User Story 38: Admin Broadcast Announcements
- **Nathan:**
 - User Story 72: Club Role Management
 - User Story 83: Message Reactions
 - User Story 74: Recurring Events
- **Rohan:**
 - User Story 11: Notification Preferences
 - User Story 48: Exam Dates Tracking
 - User Story 59: Bookmark AI Responses
- **Sheehan:**
 - User Story 86: Public Profile Views
 - User Story 7: Profile Visibility Toggle
 - User Story 14: Weekly Study Time Analytics